THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student draws the Moon at the same time each night and sees the bright part grows. The Moon is: 1

(a) waning **(b)** waxing **(c)** new **(d)** missing

2. A saturated solution of salt at 30 °C contains 36 g salt in 100 g water. In 50 g water, the maximum salt that dissolves is: 1

(a) 72 g **(b)** 18 g **(c)** 36 g **(d)** 9 g

3. Two groups of seeds are kept: Group A in water, group B dry. Only A sprouts. What is the conclusion? 1

(a) dry seeds are dead **(b)** water is needed for germination
(c) sprouting is random **(d)** seeds do not need water

4. Earthworms help soil by: 1

(a) removing humus **(b)** loosening and mixing it
(c) making it hard **(d)** polluting it

5. The weight of the same object on the Moon compared with Earth is: 1

(a) equal **(b)** one sixth **(c)** six times **(d)** zero

6. Why does purifying water require both filtration and chlorination or boiling? 1

(a) boiling removes sand **(b)** sedimentation removes germs
(c) filtration cannot remove all germs **(d)** filtration removes salt

7. A tadpole's tail disappears as it grows. This is an example of: 1

(a) change in body during growth **(b)** death
(c) response to light **(d)** reproduction

8. Which of the following resource-use decisions is sustainable? 1

(a) burning forests **(b)** planting a tree for every tree cut
(c) cutting without planting **(d)** wasting water

9. Which observation proves that the Earth rotates? 1

(a) stars appear to move across the night sky
(b) planets are bright
(c) the Sun is hot
(d) the Moon is round

10. Which of these will take the longest to settle in water? 1

(a) gravel **(b)** sand **(c)** fine clay **(d)** pebbles

11. Which pair of observations would suggest that a dry-looking object is a living seed? 1

(a) it swells and sprouts in water **(b)** it is brown and small
(c) it breaks when pressed **(d)** it is hard and dry

12. Which of the following correctly shows the relative abundance in air? 1

(a) nitrogen > oxygen > carbon dioxide **(b)** oxygen > nitrogen > carbon dioxide
(c) carbon dioxide > oxygen > nitrogen **(d)** nitrogen = oxygen

13. If the new moon occurs on 1 March, the next new moon will be about: 1

(a) 30 March **(b)** 15 March **(c)** 8 March **(d)** 1 April + 5 days

14. Using the information above, if 25 g of salt is added to 50 g water at 30 °C, the extra undissolved salt is: 1

(a) 7 g **(b)** 18 g **(c)** 25 g **(d)** 0 g

15. In the experiment above, what is the controlled variable? 1

(a) type of seeds and temperature **(b)** dry state
(c) water **(d)** germination

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do dead leaves carry out no life process? 2

17. Explain why we say forests are the lungs of the Earth. 2

18. Why do planets appear to move against the background of stars? 2

19. How do we remove insoluble impurities from honey? 2

20. A student says 'all moving things are living'. Give two counterexamples. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how the Sun's gravity keeps planets in orbit and what would happen if it vanished. 3

22. How does a cotton cloth help separate curd from whey? 3

23. Give three examples of how living things depend on each other. 3

24. Explain why hilly areas need special care for erosion. 3

OR

Explain with a diagram in words how the phases occur for the first quarter Moon. 3

25. At 20 °C, 36 g of salt dissolves in 100 g of water. How much water is needed to dissolve 90 g of salt? 3

OR

Design an experiment to show that germinating seeds produce heat. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain how a solar salt pan produces salt in stages and how impurities are removed. 5

OR

Discuss the importance of responses to stimuli in the survival of plants and animals with five examples. 5

27. Discuss with examples how our daily lifestyle can be made more 'earth-friendly'. 5

OR

Explain, using the Earth-Moon-Sun positions, why the same Moon phase can be seen all over the world on the same day but at different times. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND